```
# Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.feature_selection import SelectKBest, chi2
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
```

```
df = pd.read_csv("Churn_Modelling.csv")
df.head()
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActive
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	

```
print(df.isnull().sum())
```

```
RowNumber          0
CustomerId         0
Surname            0
CreditScore        0
Geography          0
Gender             0
Age                0
Tenure             0
Balance            0
NumOfProducts      0
HasCrCard          0
IsActiveMember     0
EstimatedSalary    0
Exited             0
dtype: int64
```

```
df.drop(['RowNumber', 'CustomerId', 'Surname'], axis=1, inplace=True)
```

```
numerical_cols = ['CreditScore', 'Age', 'Balance', 'EstimatedSalary']

for col in numerical_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    df = df[(df[col] >= lower_bound) & (df[col] <= upper_bound)]

print("Shape after removing outliers:", df.shape)
```

```
Shape after removing outliers: (9626, 11)
```

```
X = df.drop("Exited", axis=1)
y = df["Exited"]
```

```
# Label Encoding for Gender
le = LabelEncoder()
X['Gender'] = le.fit_transform(X['Gender'])
```

```
# One-Hot Encoding for Geography
X = pd.get_dummies(X, columns=['Geography'], drop_first=True)
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Convert back to DataFrame
X_scaled = pd.DataFrame(X_scaled, columns=X.columns)
X_scaled.head()
```

	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Geography
0	-0.330309	-1.097157	0.479282	-1.044601	-1.226774	-0.914163	0.646922	0.992961	0.021480	-
1	-0.444790	-1.097157	0.365625	-1.390859	0.116386	-0.914163	-1.545780	0.992961	0.216090	-
2	-1.547974	-1.097157	0.479282	1.032946	1.332056	2.529285	0.646922	-1.007089	0.240239	-
3	0.502283	-1.097157	0.138311	-1.390859	-1.226774	0.807561	-1.545780	-1.007089	-0.109299	-
4	2.073800	-1.097157	0.592939	-1.044601	0.784746	-0.914163	0.646922	0.992961	-0.365608	-


```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Convert back to DataFrame
X_scaled = pd.DataFrame(X_scaled, columns=X.columns)
X_scaled.head()
```

	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Geography
0	-0.330309	-1.097157	0.479282	-1.044601	-1.226774	-0.914163	0.646922	0.992961	0.021480	-
1	-0.444790	-1.097157	0.365625	-1.390859	0.116386	-0.914163	-1.545780	0.992961	0.216090	-
2	-1.547974	-1.097157	0.479282	1.032946	1.332056	2.529285	0.646922	-1.007089	0.240239	-
3	0.502283	-1.097157	0.138311	-1.390859	-1.226774	0.807561	-1.545780	-1.007089	-0.109299	-
4	2.073800	-1.097157	0.592939	-1.044601	0.784746	-0.914163	0.646922	0.992961	-0.365608	-


```
selector = SelectKBest(score_func=chi2, k=8)
X_selected = selector.fit_transform(X, y)

selected_columns = X.columns[selector.get_support()]
print("Selected Features:")
print(selected_columns)
```

```
Selected Features:
Index(['CreditScore', 'Gender', 'Age', 'Balance', 'IsActiveMember',
       'EstimatedSalary', 'Geography_Germany', 'Geography_Spain'],
      dtype='object')
```

```
pca = PCA(n_components=5)
X_pca = pca.fit_transform(X_scaled)

print("Explained Variance Ratio:")
print(pca.explained_variance_ratio_)

print("Total Variance Retained:", sum(pca.explained_variance_ratio_))
```

```
Explained Variance Ratio:
[0.15101401 0.10267437 0.09588078 0.09357746 0.09180924]
Total Variance Retained: 0.5349558590365022
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)

print("Training Shape:", X_train.shape)
print("Testing Shape:", X_test.shape)
```

```
Training Shape: (7700, 11)
Testing Shape: (1926, 11)
```

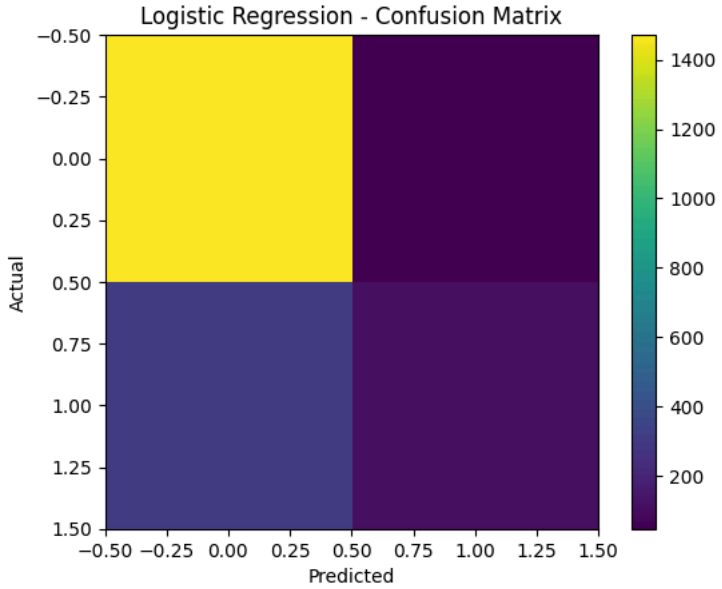
```
print("Final Dataset Shape:", X_scaled.shape)
```

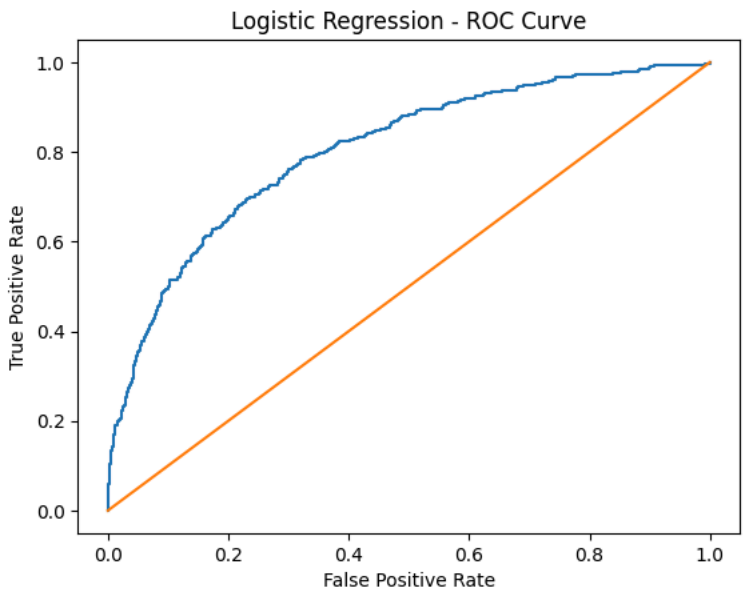
```
Final Dataset Shape: (9626, 11)
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.linear_model import LogisticRegression, LinearRegression
from sklearn.tree import DecisionTreeClassifier, DecisionTreeRegressor
from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    roc_auc_score, confusion_matrix, roc_curve,
    mean_absolute_error, mean_squared_error, r2_score
)

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.linear_model import LogisticRegression, LinearRegression
from sklearn.tree import DecisionTreeClassifier, DecisionTreeRegressor
from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor, VotingClassifier

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    roc_auc_score, confusion_matrix
)
```

```
log_clf = LogisticRegression(max_iter=1000)
log_clf.fit(X_train, y_train)

y_pred = log_clf.predict(X_test)
y_prob = log_clf.predict_proba(X_test)[:,1]

print("Logistic Regression Results")
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
print("ROC-AUC:", roc_auc_score(y_test, y_prob))

cm = confusion_matrix(y_test, y_pred)
print(cm)
plt.figure()
plt.imshow(cm)
plt.title("Logistic Regression - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.colorbar()
plt.show()

fpr, tpr, _ = roc_curve(y_test, y_prob)
plt.figure()
plt.plot(fpr, tpr)
plt.plot([0,1],[0,1])
plt.title("Logistic Regression - ROC Curve")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.show()
```

```
Logistic Regression Results
Accuracy: 0.8177570093457944
Precision: 0.6842105263157895
Recall: 0.25552825552825553
F1 Score: 0.37209302325581395
ROC-AUC: 0.8030386601815174
[[1471   48]
 [ 303  104]]
```

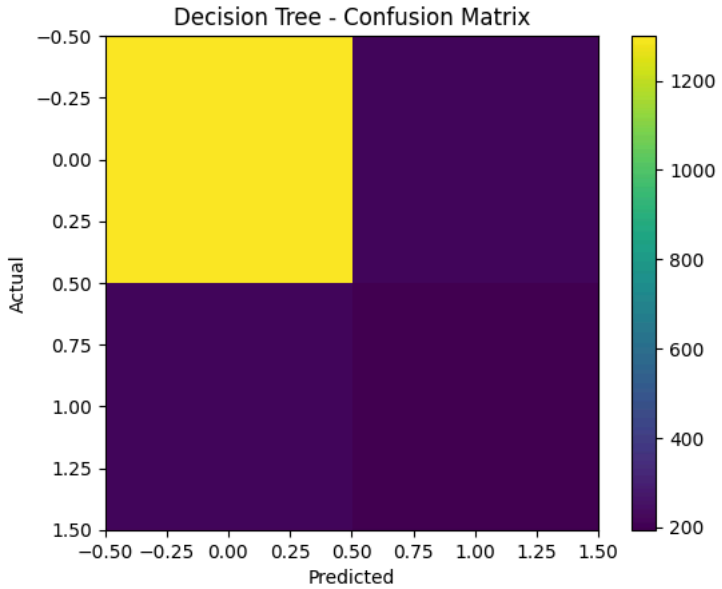


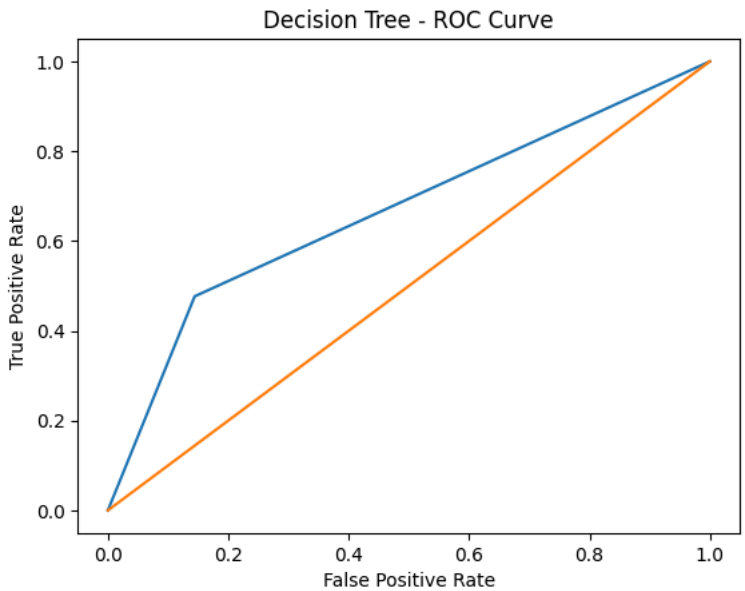



```
dt_clf = DecisionTreeClassifier(random_state=42)
dt_clf.fit(X_train, y_train)

y_pred = dt_clf.predict(X_test)
y_prob = dt_clf.predict_proba(X_test)[:,1]

print("Decision Tree Results")
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
print("ROC-AUC:", roc_auc_score(y_test, y_prob))

cm = confusion_matrix(y_test, y_pred)
print(cm)
plt.figure()
plt.imshow(cm)
plt.title("Decision Tree - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.colorbar()
plt.show()

fpr, tpr, _ = roc_curve(y_test, y_prob)
plt.figure()
```

```
plt.plot(fpr, tpr)
plt.plot([0,1],[0,1])
plt.title("Decision Tree - ROC Curve")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.show()
```

```
Decision Tree Results
Accuracy: 0.7757009345794392
Precision: 0.46973365617433416
Recall: 0.47665847665847666
F1 Score: 0.47317073170731705
ROC-AUC: 0.666242339053399
[[1300  219]
 [ 213  194]]
```

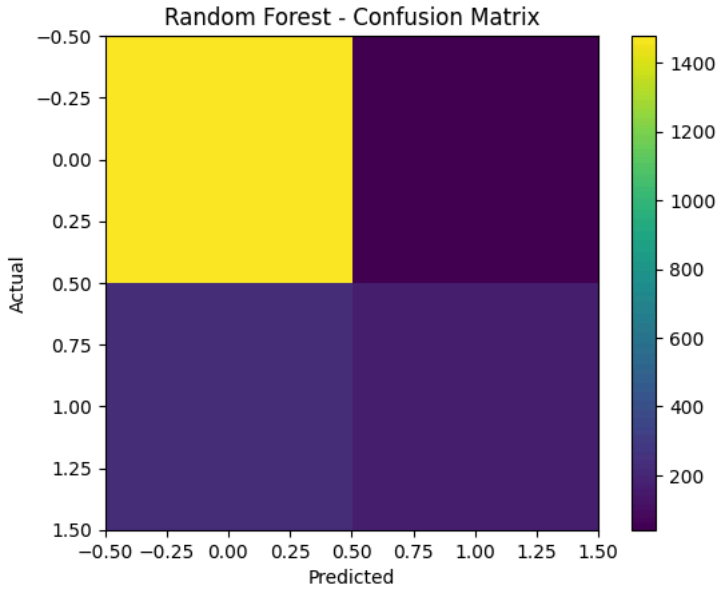


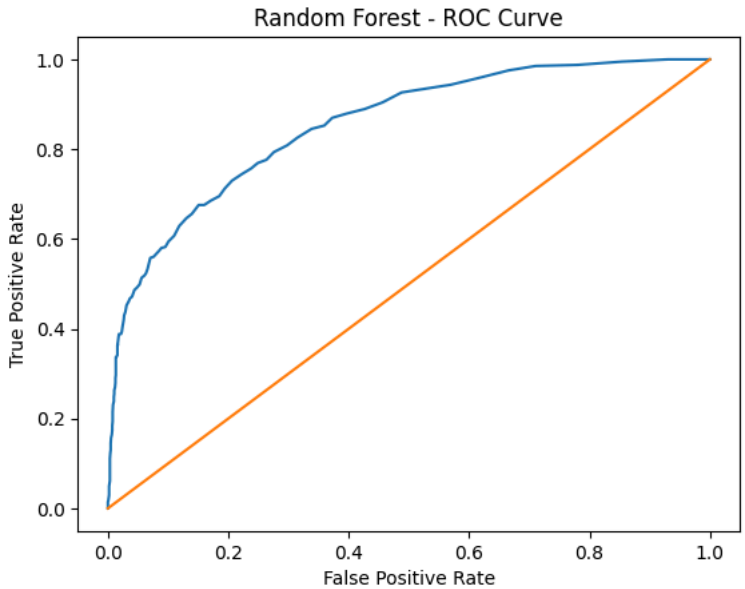



```
rf_clf = RandomForestClassifier(random_state=42)
rf_clf.fit(X_train, y_train)

y_pred = rf_clf.predict(X_test)
y_prob = rf_clf.predict_proba(X_test)[:,1]

print("Random Forest Results")
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
print("ROC-AUC:", roc_auc_score(y_test, y_prob))

cm = confusion_matrix(y_test, y_pred)
print(cm)
plt.figure()
plt.imshow(cm)
plt.title("Random Forest - Confusion Matrix")
```

```
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.colorbar()
plt.show()

fpr, tpr, _ = roc_curve(y_test, y_prob)
plt.figure()
plt.plot(fpr, tpr)
plt.plot([0,1],[0,1])
plt.title("Random Forest - ROC Curve")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.show()
```

```
Random Forest Results
Accuracy: 0.8582554517133957
Precision: 0.8101851851851852
Recall: 0.42997542997543
F1 Score: 0.5617977528089888
ROC-AUC: 0.8532583993413486
[[1478   41]
 [ 232  175]]
```



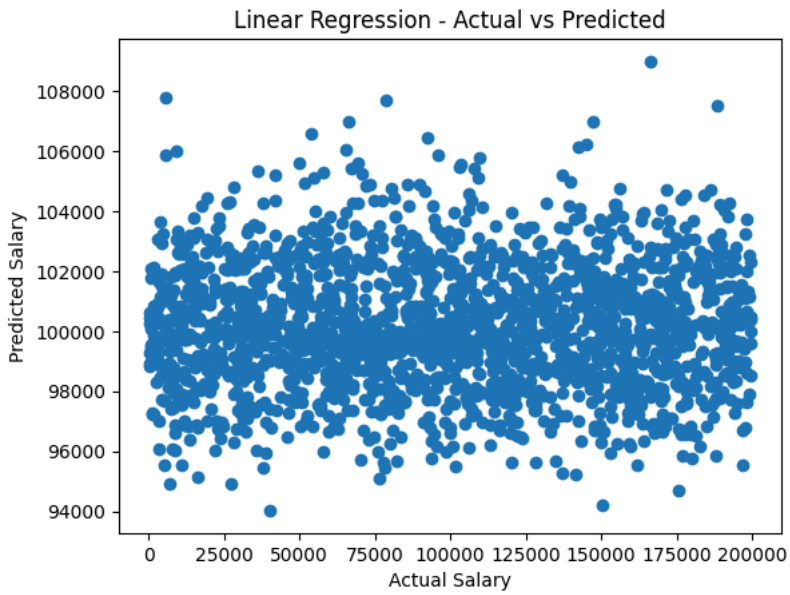



```
df_reg = df.copy()

X_reg = df_reg.drop("EstimatedSalary", axis=1)
y_reg = df_reg["EstimatedSalary"]

X_reg = pd.get_dummies(X_reg, columns=['Geography'], drop_first=True)

le = LabelEncoder()
X_reg['Gender'] = le.fit_transform(X_reg['Gender'])
```

```
scaler = StandardScaler()
X_reg_scaled = scaler.fit_transform(X_reg)

X_train_r, X_test_r, y_train_r, y_test_r = train_test_split(
    X_reg_scaled, y_reg, test_size=0.2, random_state=42
)
```

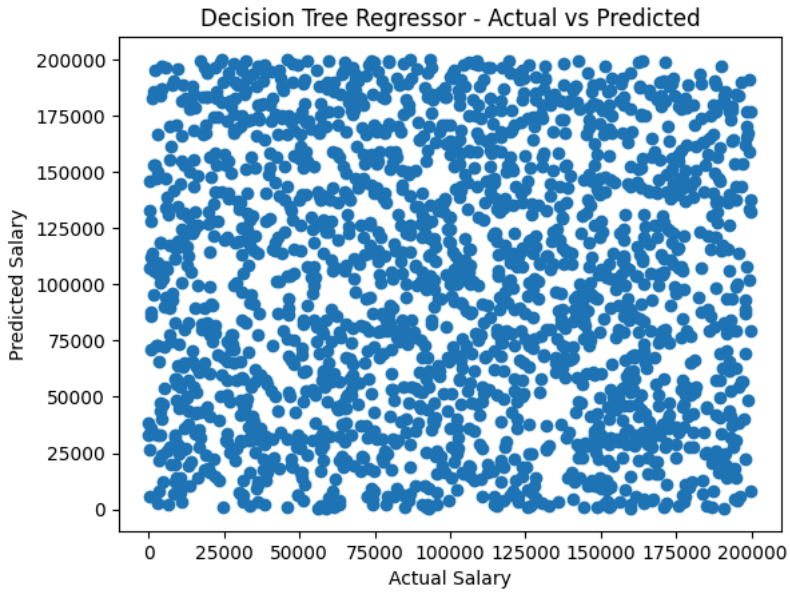
```
lin_reg = LinearRegression()
lin_reg.fit(X_train_r, y_train_r)

y_pred = lin_reg.predict(X_test_r)

print("Linear Regression Results")
print("MAE:", mean_absolute_error(y_test_r, y_pred))
print("MSE:", mean_squared_error(y_test_r, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test_r, y_pred)))
print("R2 Score:", r2_score(y_test_r, y_pred))

plt.figure()
plt.scatter(y_test_r, y_pred)
plt.title("Linear Regression - Actual vs Predicted")
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.show()
```

```
Linear Regression Results
MAE: 48864.184284980984
MSE: 3222894106.420117
RMSE: 56770.53907107204
R2 Score: -0.001991248061450568
```



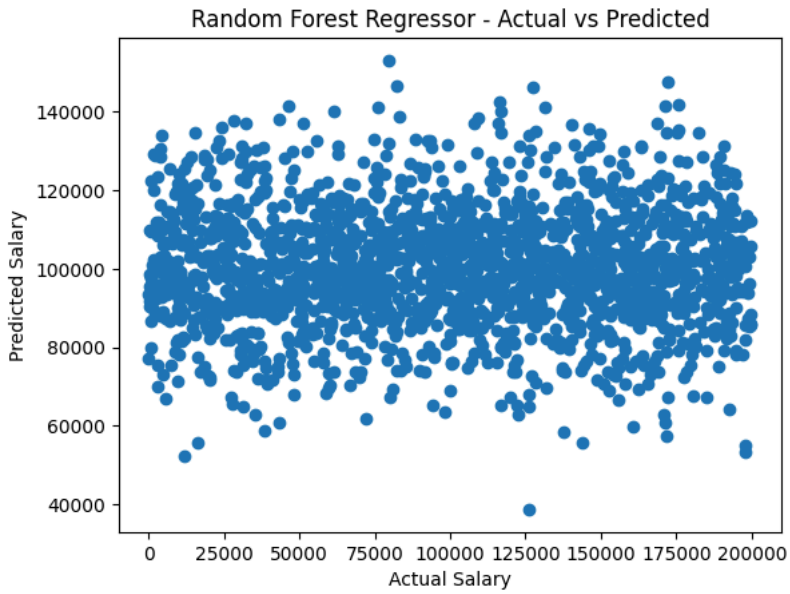

```
dt_reg = DecisionTreeRegressor(random_state=42)
dt_reg.fit(X_train_r, y_train_r)

y_pred = dt_reg.predict(X_test_r)

print("Decision Tree Regressor Results")
print("MAE:", mean_absolute_error(y_test_r, y_pred))
print("MSE:", mean_squared_error(y_test_r, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test_r, y_pred)))
print("R2 Score:", r2_score(y_test_r, y_pred))

plt.figure()
plt.scatter(y_test_r, y_pred)
plt.title("Decision Tree Regressor - Actual vs Predicted")
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.show()
```

```
Decision Tree Regressor Results
MAE: 66925.90080477673
MSE: 6774343949.023931
RMSE: 82306.40284342361
R2 Score: -1.1061298088443006
```




```
rf_reg = RandomForestRegressor(random_state=42)
rf_reg.fit(X_train_r, y_train_r)

y_pred = rf_reg.predict(X_test_r)

print("Random Forest Regressor Results")
print("MAE:", mean_absolute_error(y_test_r, y_pred))
print("MSE:", mean_squared_error(y_test_r, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test_r, y_pred)))
print("R2 Score:", r2_score(y_test_r, y_pred))

plt.figure()
plt.scatter(y_test_r, y_pred)
plt.title("Random Forest Regressor - Actual vs Predicted")
plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")
plt.show()
```

```
Random Forest Regressor Results
MAE: 50117.6550611388
MSE: 3463588715.5220294
RMSE: 58852.2617706578
R2 Score: -0.07682271438088728
```




```
from sklearn.ensemble import VotingClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score
```

```
import pandas as pd
```

```
log_clf = LogisticRegression(max_iter=1000)
dt_clf = DecisionTreeClassifier(random_state=42)
rf_clf = RandomForestClassifier(random_state=42)

log_clf.fit(X_train, y_train)
dt_clf.fit(X_train, y_train)
rf_clf.fit(X_train, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

```
models = {
    "Logistic Regression": log_clf,
    "Decision Tree": dt_clf,
    "Random Forest": rf_clf
}

results = []

for name, model in models.items():
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:,1]

    results.append([
        name,
        accuracy_score(y_test, y_pred),
        precision_score(y_test, y_pred),
        recall_score(y_test, y_pred),
        f1_score(y_test, y_pred),
        roc_auc_score(y_test, y_prob)
    ])

results_df = pd.DataFrame(results, columns=["Model","Accuracy","Precision","Recall","F1-Score","ROC-AUC"])
results_df
```

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.817757	0.684211	0.255528	0.372093	0.803039
1	Decision Tree	0.775701	0.469734	0.476658	0.473171	0.666242
2	Random Forest	0.858255	0.810185	0.429975	0.561798	0.853258

```
hard_vote = VotingClassifier(
    estimators=[
        ('lr', log_clf),
        ('dt', dt_clf),
        ('rf', rf_clf)
    ],
    voting='hard'
)

hard_vote.fit(X_train, y_train)

y_pred_hard = hard_vote.predict(X_test)

print("Hard Voting")
print("Accuracy:", accuracy_score(y_test, y_pred_hard))
print("Precision:", precision_score(y_test, y_pred_hard))
print("Recall:", recall_score(y_test, y_pred_hard))
print("F1 Score:", f1_score(y_test, y_pred_hard))

cm = confusion_matrix(y_test, y_pred_hard)
print("Confusion Matrix:")
print(cm)
```

```
Hard Voting
Accuracy: 0.8515057113187954
Precision: 0.7951219512195122
Recall: 0.4004914004914005
F1 Score: 0.5326797385620915
Confusion Matrix:
[[1477   42]
 [ 244  163]]
```

```
soft_vote = VotingClassifier(
    estimators=[
        ('lr', log_clf),
        ('dt', dt_clf),
        ('rf', rf_clf)
    ],
    voting='soft'
)

soft_vote.fit(X_train, y_train)

y_pred_soft = soft_vote.predict(X_test)
y_prob_soft = soft_vote.predict_proba(X_test)[:,1]

print("Soft Voting")
print("Accuracy:", accuracy_score(y_test, y_pred_soft))
print("Precision:", precision_score(y_test, y_pred_soft))
print("Recall:", recall_score(y_test, y_pred_soft))
print("F1 Score:", f1_score(y_test, y_pred_soft))
print("ROC-AUC:", roc_auc_score(y_test, y_prob_soft))

cm = confusion_matrix(y_test, y_pred_soft)
print("Confusion Matrix:")
print(cm)
```

```
Soft Voting
Accuracy: 0.833852544132918
Precision: 0.6548042704626335
Recall: 0.4520884520884521
F1 Score: 0.5348837209302325
ROC-AUC: 0.8301109775764154
Confusion Matrix:
[[1422   97]
 [ 223  184]]
```

```
results.append([
    "Hard Voting",
    accuracy_score(y_test, y_pred_hard),
    precision_score(y_test, y_pred_hard),
    recall_score(y_test, y_pred_hard),
    f1_score(y_test, y_pred_hard),
    "-"
])

results.append([
    "Soft Voting",
    accuracy_score(y_test, y_pred_soft),
    precision_score(y_test, y_pred_soft),
    recall_score(y_test, y_pred_soft),
    f1_score(y_test, y_pred_soft),
    roc_auc_score(y_test, y_prob_soft)
])

final_results = pd.DataFrame(results, columns=["Model","Accuracy","Precision","Recall","F1-Score","ROC-AUC"])
final_results
```

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.817757	0.684211	0.255528	0.372093	0.803039
1	Decision Tree	0.775701	0.469734	0.476658	0.473171	0.666242
2	Random Forest	0.858255	0.810185	0.429975	0.561798	0.853258
3	Hard Voting	0.851506	0.795122	0.400491	0.532680	-
4	Soft Voting	0.833853	0.654804	0.452088	0.534884	0.830111